

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration, Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

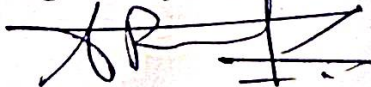
Annexure B Analytical Problem Solving

Criteria	Problem Understanding (2)	Method Selection (2)	Solution Process (2.5)	Accuracy of Calculation (2)	Interpretation of Results (1.5)	Total(10)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Bhargava Aravind (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.

Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.

To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**. To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

5. Standardization of Patient Age Data (5 Marks)

Given the following ages of insured individuals:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

- (a) Compute the **mean absolute deviation (MAD)** of age.
- (b) Compute the **z-score** for the first four observations.

Data warehousing and data mining

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①

Step 1:

$$\bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4} = \frac{17.5}{4} = 4.375$$

$$\bar{y} = \frac{15 + 18 + 25 + 20}{4} = \frac{78}{4} = 19.5$$

Step 2: Covariance Table

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125

$$\sum (x - \bar{x})(y - \bar{y}) = 12.25$$

Population Covariance

$$\text{Cov}(x, y) = \frac{12.25}{4} = 3.0625$$

Mean of x = 4.375

Mean of y = 19.5 Covariance = 3.0625

⑥ 110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

Bin 1

110, 112, 115, 115, 118, 120, 120, 122, 124

Mean:

$$\frac{1056}{9} = 117.33$$

Smoothed:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33

117.33, 117.33, 117.33

Bin 2

124, 126, 126, 126, 126, 130, 135, 135, 138

$$\frac{1166}{9} = 129.56$$

Bin 3

138, 138, 138, 140, 145, 150, 152, 160, 180

$$\frac{1341}{9} = 149$$

Smoothed:

149 (Repeated 9 times)

$$+ \text{Min} = 18.5$$

$$+ \text{Max} = 39.5$$

$$\text{Range} = 21$$

Using 3 equal-width bins

$$\text{width} = \frac{39.5 - 18.5}{3} = 7$$

Bins:

$$+ \text{Bin}_1: 18.5 - 25.5$$

$$+ \text{Bin}_2: 25.5 - 32.5$$

$$+ \text{Bin}_3: 32.5 - 39.5$$

Result

$$+ \text{Bin}_1 = 18.5, 18.5, 25.5, 25.5$$

$$+ \text{Bin}_2 = 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5$$

$$+ \text{Bin}_3 = 32.5, 32.5, 32.5, 32.5, 39.5, 39.5, 39.5, 39.5$$

$$\textcircled{A} \text{ Bin}_1: 120, 140, 145, 145, 148, 150, 150, 150, 155$$

$$\text{Median} = 148$$

Smoothed:

148 repeated 9 times

$$\text{Bin}_2: 155, 155, 155, 160, 160, 165, 170, 170, 175$$

$$\text{Median} = 160$$

Smoothed:

160 repeated 9 times

B.O. 3: 175, 175, 180, 190, 200, 210, 230, 300, 700

Median = 200

Smoothed:

200 repeated a times

⑤ Data: 18, 22, 25, 42, 28, 43, 33, 35, 36, 28

(a) Mean absolute deviation

Mean =

$$\bar{x} = \frac{330}{10} = 33$$

absolute deviations:

15, 11, 8, 9, 5, 10, 0, 2, 23, 5

Sum = 88

$$MAD = \frac{88}{10} = 8.8$$

(b) Z-score

Mean = 33

$$\sigma = \sqrt{\frac{1096}{10}} = \sqrt{109.6} \approx 10.47$$

$$Z = \frac{x - \bar{x}}{\sigma}$$

Age	Z-score
18	-1.43
22	-1.05
25	-0.76
42	0.86

} Answers
of Z-score